

Amir P. Adam

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SUMMARY

ECE graduate with practical experience in embedded firmware, IoT system integration, and edge computing. Worked with ESP32, STM32, and Raspberry Pi on real projects involving C/C++, Python, MQTT, and REST APIs. Strong debugging instincts and a track record of delivering complete, working systems.

EDUCATION

Chandigarh University

Bachelor of Engineering in Electronics & Communications

Mohali, India

Aug. 2022 – Present

- Relevant Coursework: Embedded Systems, VLSI Design, Communication Systems, Digital Signal Processing

PROJECTS

IoT Food Monitoring System | *ESP32, DHT11, MQ135, AWS IoT Core, MQTT/TLS, AWS Lambda, Firebase, React Native*

- Architected a secure edge-to-cloud pipeline: ESP32 transmits DHT11/MQ135 sensor data via TLS-encrypted MQTT to AWS IoT Core with <2s ingestion latency.
- Built a serverless AWS Lambda backend to parse, validate, and compute food freshness status; wrote real-time and historical data to Cloud Firestore and triggered FCM push alerts to React Native app.
- Delivered a cross-platform mobile dashboard with real-time Firestore sync, achieving <500ms UI update latency from sensor threshold breach to user notification.

Edge AI Surveillance System | *Raspberry Pi 5, Python, OpenCV, TensorFlow Lite, FastAPI, Telegram Bot API* Mar. 2026

- Achieved ~15 FPS real-time inference on Raspberry Pi 5 using quantized MobileNet SSD, maintaining >85% detection confidence for human targets.
- Reduced alert latency to <2s via multithreaded asynchronous Telegram notifications with annotated snapshots and metadata.
- Served live MJPEG streams and logged 100+ detection events in SQLite with timestamps via a FastAPI web interface.

Disaster-Resilient Offline Mesh Communication System | *ESP32, LoRa*

Oct. – Nov. 2025

- Established decentralized peer-to-peer messaging across >500m line-of-sight range using ESP32 + LoRa without cellular or internet infrastructure.
- Cut active power consumption by ~35% via optimized duty-cycling and deep-sleep firmware, extending battery runtime in emergency scenarios.
- Achieved <3s hop-to-hop relay latency for situational awareness packets with integrated local sensor acquisition.

TECHNICAL SKILLS

Programming: C/C++, Python, TypeScript

Embedded Systems: ESP32, STM32, Raspberry Pi, Sensor Integration, I2C, SPI, UART

Cloud & IoT: AWS IoT Core, AWS Lambda, Firebase Firestore & FCM, MQTT/TLS, REST APIs

AI & Vision: TensorFlow Lite, OpenCV, MobileNet SSD, Edge Inference

Tools & Platforms: STM32CubeIDE, PlatformIO, Arduino IDE, VSCode, Git, GitHub, FastAPI, React Native

LANGUAGES

Arabic: Native (Sudanese Dialect) **English:** Proficient (C1)